Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Portfolio | Virginia, USA

PROFESSIONAL SUMMARY

Software engineer with 8+ years owning production distributed systems at scale - including a RabbitMQ-based async processing system with retry logic and idempotency guarantees, multi-tenant PostgreSQL schemas across 100K+ monthly active users, and Libsodium-encrypted AWS S3 data pipelines at 100% PHI compliance. Hands-on agentic AI systems: Leasel is an 8-agent LangGraph orchestration runtime with durable state management, trace-level observability, and conditional HITL escalation; AgentProbe is an evaluation infrastructure with golden datasets, regression harnesses, and offline + online evaluation loops. Built hybrid search with citation grounding over 1,240 residential lease PDFs - 93.6% precision, 87.9% F1. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). AgentProbe: github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Agentic Orchestration Runtime | LangGraph Multi-Agent Systems | RAG + Hybrid Search + Citation Grounding | Evaluation Infrastructure (Golden Datasets + Regression Harnesses) | RabbitMQ Async Processing | Durable State + HITL Escalation | Trace-Level Observability | Multi-Tenant Backend Engineering

WORK EXPERIENCE

Entrata (PropTech / Real Estate SaaS)

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Owned the RabbitMQ-based async processing system** for gross-rent updates across thousands of tenant records - designed retry logic and idempotency guarantees that eliminated double-processing failures; owned the system through production incidents and annual rent-schedule changes across a multi-tenant platform serving 100K+ monthly active users.
- **Optimized 200+ SQL queries (30% performance gain)** through execution plan analysis, indexing, partitioning, and stored procedure refactoring; owned complex multi-tenant PostgreSQL schemas for eligibility, financial, and document workflows. A fast query returning incorrect regulatory data is a production failure - speed and correctness are both first-class requirements.
- **Owned the annual HUD income-limit import pipeline for 7 years** - a few thousand federal source rows expanded into ~180,000 domain records powering eligibility, rent, and tax-credit calculations. Every release included baseline validation, reconciliation, and rollback preparation. Zero compliance violations across the ownership period.
- **Led the Affordable Housing module integration for the AWS S3 document migration** - Libsodium-encrypted pipelines achieving 100% PHI compliance on all resident data (zero security incidents); validated migrated content and confirmed customer-facing workflows remained functional after cutover.
- **Shipped 20+ full-stack production modules** (Python, PostgreSQL, RabbitMQ, JavaScript) across 120+ Agile sprints; led microservices redesign using Docker and Kubernetes - 50% scalability improvement, 40% deployment time reduction; eliminated 80% of manual operational workload through Python/Bash automation.

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built an agentic mission-planning orchestration runtime using LangGraph (parse → plan → retrieve → validate → conditional routing) with durable state management - delivering a risk-aware compliance strategy with grounded rejection reasons for non-compliant missions.
- Implemented a RAG layer with hybrid search over FAA Part 107 + operational-guidance corpus, answering operator queries with citation grounding to source regulation; every compliance check functions as a hypothesis test.
- Engineered validation guardrails enforcing altitude limits, battery budgets, and geometric no-fly-zone checks - blocking non-compliant missions before export.

George Mason University

Aug 2024 - Aug 2025

Research Associate · College of Engineering and Computing

- Built multi-source prediction models (Python, R) on WHO and UK Road Safety datasets - $R^2 = 0.977$; applied Isolation Forest to automate data integrity checks across 3 research pipelines.
- Fine-tuned a custom classification model using HuggingFace Transformers and PyTorch; deployed inference API on Quart; presented model design, training methodology, and offline evaluation results to faculty.

Entrata India, Pune

Jul 2017 - Sep 2018

Associate Software Engineer

- Built property management modules (Python, Java, PostgreSQL) including income limits import (1.6M records), cloud migration to AWS S3, and compliance workflows for affordable housing certification.

PROJECTS

Leasel: Agentic AI Platform - Orchestration Runtime + RAG + Evaluation

LangGraph · FastAPI · TF-IDF · OCR · SQLite

Built a three-surface enterprise AI platform sharing one evidence model: (1) **Document Intelligence** - OCR ingestion, section-aware chunking, hybrid search with TF-IDF retrieval, citation grounding to exact source clauses, missing/conflicting clause detection, human-review gate on regulated decisions; (2) **Support Investigation** - 8-agent LangGraph orchestration runtime with durable state (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) producing ranked, evidence-backed root-cause hypotheses with conditional HITL escalation; (3) **Tenant Self-Service** - eligibility questions always route to staff, never resolved by AI. Core design rule enforced in code: LLM reasons; deterministic or human logic makes every final regulated call. **Results:** 93.6% precision, 87.9% F1 across 10 compliance rules on 1,240 residential lease PDFs (5 US states). github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe: Open-Source AI Evaluation Infrastructure

Python · FastAPI · LangGraph · ChromaDB · CI/CD

Built a fully local, zero-cost evaluation infrastructure with a 15-endpoint FastAPI backend, 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression harness with golden datasets, model routing across OpenAI / Anthropic / Groq / Ollama, and offline + online evaluation loops - used to evaluate Leasel across 1,240+ compliance test cases. github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · Docker · BERT

Reduced manual clinical retrieval time ~60% by building a Flask + React NLP backend translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings; enforced HIPAA-compliant RBAC across structured medical records.
github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur — 2017

SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, PL/SQL, R, Bash

AI & Agents: LangChain, LangGraph, Orchestration Runtime, Durable State Management, Multi-Step Reasoning, HITL, Qdrant, ChromaDB, RAG Pipelines, Hybrid Search, Citation Grounding, Golden Datasets, Regression Harnesses, Offline Evaluation, Online Evaluation, LLM Evaluation, LLM-as-Judge, Model Routing, Hallucination Detection, LLMOps, Trace-Level Observability, Agent Tracing, Prompt Engineering, OpenAI API, Anthropic API, Groq, Ollama, FastAPI, MCP

ML & Data: PyTorch, HuggingFace, Scikit-learn, XGBoost, Isolation Forest, Random Forest, Feature Engineering, Pandas, NumPy, spaCy, BERT

Infrastructure: Azure, AWS (S3, EC2, RDS, Glue), Docker, Kubernetes, Prometheus, Grafana, OpenTelemetry, RabbitMQ, PostgreSQL, Solr, Redis, CI/CD (GitHub Actions), Microservices, Multi-Tenant Architecture, Data Isolation, Distributed Systems